

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI**Work Integrated Learning Programmes Division**

Cluster Programme - M. Tech in AI & ML

II Semester , 2023 – 24(July,2024)

Mid semester Examination (**MAKEUP**)

Course No : AIMLC ZC418
 Course Title : Introduction to Statistical Methods
 Nature of Exam. : Open Book (Online)
 Weightage : 30 Marks
 Duration : 120 minutes
 Date : 27th July,2024_10 AM

*Number of questions:8**Number of Pages: 2*

Q. No	Question	Marks
Q.1	Let A and B are two independent events with probabilities 0.25 and 0.20 respectively. Then find the following a) $P(A \cup B)$ b) $P(A \cap B)$ c) $P(A/B)$ d) $P(B/A)$ e) $P(A^c/B)$ f) $P(A^c \cap B^c)$ (where A^c indicates the compliment of event A)	3M
Q.1 Ans	Solution: a) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ $= 0.25 + 0.20 - 0.05$ (since $P(A \cap B) = P(A)P(B)$ for independent events) $= 0.40$ (0.5 mark) b) $P(A \cap B) = P(A)P(B)$ (since A and B are independent) $= 0.25 \times 0.20$ $= 0.05$ (0.5 mark) c) $P(A/B) = P(A \cap B) / P(B)$ (definition of conditional probability) $= 0.05 / 0.20$ $= 0.25$ (0.5 mark) d) $P(B/A) = P(A \cap B) / P(A)$ (definition of conditional probability) $= 0.05 / 0.25$ $= 0.20$ (0.5 mark) e) $P(A^c/B) = 1 - P(A/B)$ (since A^c is the complement of A) $= 1 - 0.25$ $= 0.75$ (0.5 mark) f) $P(A^c \cap B^c) = P(A^c)P(B^c)$ (since A^c and B^c are independent) $= (1 - 0.25)(1 - 0.20)$ $= 0.75 \times 0.80$ $= 0.60$ (0.5 mark)	
Q.2	If mean and standard deviation of a binomial distribution are given as 4 and $(2/\sqrt{3})$. Then find a) The probability distribution function	4M

	b) Expectation of X^2 c) $P(6 < X < 9)$																																								
Q2. Ans	Given $\mu = np = 4$ and $\sigma = \sqrt{np(1 - p)} = 2/\sqrt{3}$ Gives $p = 2/3$ and $n = 6$ (1mark) a) The probability distribution function: $P(X = k) = \binom{6}{k} \left(\frac{2}{3}\right)^k \left(\frac{1}{3}\right)^{6-k}$ where $k = 0, 1, 2, 3, 4, 5, 6$ (1mark) b) Expectation of X^2 : As $E(X^2) = \text{Var}(X) + [E(X)]^2$ $= \frac{4}{3} + 4^2 = \frac{4}{3} + 16 = \frac{4}{3} + \frac{48}{3} = \frac{52}{3}$ (1mark) c) $P(6 < X < 9)$: Since X is a binomial random variable with $n = 6$, the maximum value X can take is 6. Therefore, $P(6 < X < 9) = 0$, as there are no values of X that lie between 6 and 9. (1mark)																																								
Q.3	<table><tr><th>Vehicle</th><th>Credit card/Debit card</th><th>Travel</th><th>Employment</th></tr><tr><td>BMW</td><td>Credit card</td><td>Flight</td><td>Business</td></tr><tr><td>Scooter</td><td>Debit card</td><td>Bus</td><td>Salaried</td></tr><tr><td>Bicycle</td><td>None</td><td>Local train</td><td>Business</td></tr><tr><td>Verna</td><td>Credit card</td><td>Vande Bharat</td><td>Salaried</td></tr><tr><td>Scooter</td><td>Credit card</td><td>Flight</td><td>Salaried</td></tr><tr><td>Auto</td><td>None</td><td>Bus</td><td>Business</td></tr><tr><td>BMW</td><td>Debit card</td><td>Vande Bharat</td><td>Salaried</td></tr><tr><td>No</td><td>No</td><td>Bus</td><td>Salaried</td></tr></table>	Vehicle	Credit card/Debit card	Travel	Employment	BMW	Credit card	Flight	Business	Scooter	Debit card	Bus	Salaried	Bicycle	None	Local train	Business	Verna	Credit card	Vande Bharat	Salaried	Scooter	Credit card	Flight	Salaried	Auto	None	Bus	Business	BMW	Debit card	Vande Bharat	Salaried	No	No	Bus	Salaried	Based on the data is it possible to find whether the following is salaried or business person. If possible find with the help of suitable theorem / algorithm. Show the working, mention assumptions if any. "Vehicle = scooter; Credit Card = yes, Travel = Bus"			8M
Vehicle	Credit card/Debit card	Travel	Employment																																						
BMW	Credit card	Flight	Business																																						
Scooter	Debit card	Bus	Salaried																																						
Bicycle	None	Local train	Business																																						
Verna	Credit card	Vande Bharat	Salaried																																						
Scooter	Credit card	Flight	Salaried																																						
Auto	None	Bus	Business																																						
BMW	Debit card	Vande Bharat	Salaried																																						
No	No	Bus	Salaried																																						

Q.3
Ans

Let

- S = Salaried
- B = Business
- V = Vehicle (scooter)
- C = credit card (yes)
- T = Travel (Bus)

②

$$P(S | V, C, T) = \frac{P(V, C, T | S) P(S)}{P(V, C, T)}$$

$$P(B | V, C, T) = \frac{P(V, C, T | B) P(B)}{P(V, C, T)}$$

②

Total Salaried = 5
Business = 3

vehicle = scooter, Salaried = 2, Business = 0
Instance = credit card, Salaried = 3, Business = 1
Instance travel = Bus, Salaried = 3, Business = 2

①

$$P(S) = \frac{5}{8}$$

$$P(B) = \frac{3}{8}$$

Total Probability $P(V, C, T)$

There is no direct instance for the combination given (scooter, yes, Bus), therefore both probabilities are zero.

③

We cannot determine whether the person is salaried or Business based on given data.

Q.4

Let X be a discrete random variable whose probability distribution function is given as $p(x) = k(x+1)/4$ where $x = -1, 0, 1, 2$, then find

- a) k value
- b) Expectation of x
- c) Expectation of x^2
- d) Variance
- e) $P(-1 < x < 2)$

4M

Q4.
Ans

$$p(x) = \frac{k(x+1)}{4} \quad \text{for } x = -1, 0, 1, 2$$

(a): Find the value of (k): $\sum_x p(x) = 1$

$$p(-1) + p(0) + p(1) + p(2) = 1$$

$$p(-1) = \frac{k(-1+1)}{4} = 0$$

$$p(0) = \frac{k(0+1)}{4} = \frac{k}{4}$$

$$p(1) = \frac{k(1+1)}{4} = \frac{2k}{4} = \frac{k}{2}$$

$$p(2) = \frac{k(2+1)}{4} = \frac{3k}{4}$$

$$0 + \frac{k}{4} + \frac{k}{2} + \frac{3k}{4} = 1$$

$$\frac{6k}{4} = 1$$

$$k = \frac{2}{3}$$

-----→1Mark

(b): Expectation of (x)

$$E(x) = \sum_x x \cdot p(x)$$

$$E(x) = (-1) \cdot p(-1) + 0 \cdot p(0) + 1 \cdot p(1) + 2 \cdot p(2)$$

$$E(x) = (-1) \cdot 0 + 0 \cdot \frac{2}{12} + 1 \cdot \frac{2}{6} + 2 \cdot \frac{1}{2}$$

$$E(x) = 0 + 0 + \frac{1}{3} + 1$$

$$E(x) = \frac{4}{3}$$

-----→1Mark

(c): Expectation of (x²):

$$E(x^2) = \sum_x x^2 \cdot p(x)$$

$$E(x^2) = (-1)^2 \cdot p(-1) + 0^2 \cdot p(0) + 1^2 \cdot p(1) + 2^2 \cdot p(2)$$

$$E(x^2) = 1 \cdot 0 + 0 \cdot \frac{2}{12} + 1 \cdot \frac{1}{3} + 4 \cdot \frac{1}{2}$$

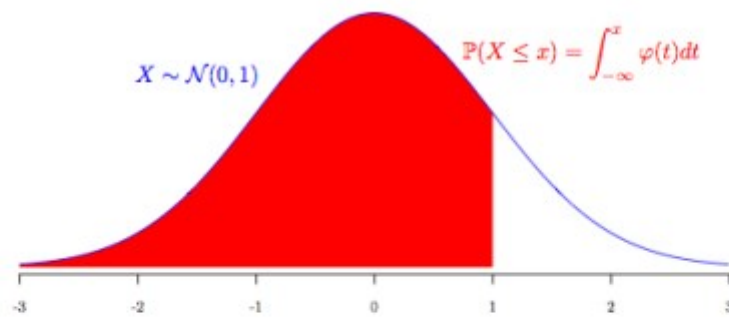
$$E(x^2) = 0 + 0 + \frac{1}{3} + 2$$

	$E(x^2) = \frac{7}{3}$ -----→ 1Mark (d): Variance: $\text{Var}(x) = E(x^2) - [E(x)]^2$ $\text{Var}(x) = \frac{7}{3} - \left(\frac{4}{3}\right)^2$ $\text{Var}(x) = \frac{5}{9}$ -----→ $\frac{1}{2}$ Mark (e): Probability $P(-1 < x < 2)$: The values of (x) within the interval $(-1 < x < 2)$ are 0 and 1. $P(-1 < x < 2) = p(0) + p(1)$ $P(-1 < x < 2) = \frac{k}{4} + \frac{k}{2}$ $P(-1 < x < 2) = \frac{2}{3} \cdot \frac{1}{4} + \frac{2}{3} \cdot \frac{1}{2}$ $P(-1 < x < 2) = \frac{1}{6} + \frac{1}{3}$ $P(-1 < x < 2) = \frac{1}{2}$ -----→ $\frac{1}{2}$ Mark	
Q.5	Consider the probability density function $f(X,Y) = k (X + Y)/8$ where $0 < X < 2$ and $0 < Y < 2$. Then find - k value - Marginal probability density function of X - Marginal probability density function of Y - Are X and Y are independent? Justify it	4M
Q.5 Ans	$\int_0^2 \int_0^2 \frac{k(x+y)}{8} dx dy = 1,$ Solving this with proper steps yield K =1 (1Mark) $f(x) = \int_0^2 \frac{x+y}{8} dy = \frac{x+1}{4}$ (1Mark) $f(y) = \int_0^2 \frac{x+y}{8} dx = \frac{y+1}{4}$ (1Mark) $f(xy) \neq f(x).f(y)$. Hence not independent (1Mark)	
Q.6	Salaries of 500 employees in a company follows normal distribution with an average salary of 20(lakhs per annum) and variance of 4(lakhs).	4M

	Then find the number of employees - whose salary is above 22(lakhs) - whose salary is below 16(lakhs) - Whose salary is between 16 and 22(Lakhs) - Are there any inconsistencies observed in this data?	
Q.6 Ans	a) Employees with salary above 22 lakhs - Calculate the Z-score for 22 lakhs: $Z = \frac{22-20}{2} = 1$ - Find the probability corresponding to $Z = 1$ using the standard normal distribution table. The probability that a Z-score is less than 1 is approximately 0.8413. - The probability of a salary being above 22 lakhs is: $P(X > 22) = 1 - P(Z < 1) = 1 - 0.8413 = 0.1587$ - Number of employees with a salary above 22 lakhs: $0.1587 \times 500 = 79.35 \approx 79$ b) Employees with salary below 16 lakhs - Calculate the Z-score for 16 lakhs: $Z = \frac{16-20}{2} = -2$ - Find the probability corresponding to $Z = -2$. The probability that a Z-score is less than -2 is approximately 0.0228. - Number of employees with a salary below 16 lakhs: $0.0228 \times 500 = 11.4 \approx 11$ c) Employees with salary between 16 and 22 lakhs - The probability of a salary being between 16 and 22 lakhs is: $P(16 < X < 22) = P(Z < 1) - P(Z < -2) = 0.8413 - 0.0228 = 0.8185$ - Number of employees with a salary between 16 and 22 lakhs: $0.8185 \times 500 = 409.25 \approx 409$ d) Checking for inconsistencies Let's sum the number of employees calculated: - Employees with salary above 22 lakhs: 79 - Employees with salary below 16 lakhs: 11 - Employees with salary between 16 and 22 lakhs: 409 Total = $79 + 11 + 409 = 499$ The total number of employees is 500, but our calculations sum to 499. This slight discrepancy is due to rounding errors during the calculations. Therefore, there is no significant inconsistency in the data.	1M 1M 1M 1M
Q.7	- Consider a population which follows binomial distribution. Sampling is done with a sample of size 45 and the sample mean is 10 with standard deviation 2. Can we conclude that the sampling distribution of mean also follows binomial distribution? Support your answer. - Consider a population which follows binomial distribution. Sampling is done with a sample of size 25 and the sample mean is 10 with standard deviation 2. Can we conclude that the sampling distribution of mean also follows binomial distribution? Support your answer.	3M

	c) Consider a population which normal distribution. Sampling is done with a sample of size 9. Then discuss about the distribution that sampling distribution of mean follows.	
	Solution: a) For Binomial distribution $\mu = np$ and $\sigma^2 = npq$. In given problem $n = 45$ and sample mean $\bar{x} = 10$ and S.D. = 2. Suppose sample statistics follow the binomial distribution then $\bar{x} = np = 10 \rightarrow 10 = p \cdot 45$. This gives $p = \frac{10}{45} = \frac{2}{9}$. For this value of p the S.D. = $\sqrt{45 \times \frac{2}{9} \times \left(1 - \frac{2}{9}\right)} = 2.79$. But given S.D is 2. So sampling distribution of mean does not follow binomial distribution. (1 Mark) b) Here, $\bar{x} = np = 10 \rightarrow 10 = p \cdot 25$. This gives $p = \frac{10}{25} = \frac{2}{5}$. For this value of p the S.D. = $\sqrt{25 \times \frac{2}{5} \times \left(1 - \frac{2}{5}\right)} = 2.45$. But given S.D is 2. So sampling distribution of mean does not follow the binomial distribution. (1 Mark) c) The Central Limit Theorem states that if the population itself is normally distributed, the sampling distribution of the sample mean will also be normally distributed, regardless of the sample size. Hence the sampling distribution follows the normal distribution. (1 Mark)	

z	0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414
-0.1	.46017	.45620	.45224	.44828	.44433	.44034	.43640	.43251	.42858	.42465
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-1	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08692	.08534	.08379	.08226
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-2	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-3	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-4	.00003	.00003	.00003	.00003	.00003	.00003	.00002	.00002	.00002	.00002



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990